Insurance Portfolio Analysis – Key Insights

Portfolio Size

The dataset contains approximately one million insurance policies, creating a realistic environment for analyzing large-scale insurance portfolios.

Premium Revenue

The company generated approximately ₹24.01 Crores in premium revenue from policies sold in 2024.

Claims Cost

Total claim payouts reached approximately ₹98.81 Crores, significantly exceeding the premium revenue.

Claim Frequency

The claim frequency is approximately 4.94%, meaning around 5 out of every 100 policies result in a claim.

Average Claim Amount

The average claim amount is approximately ₹20,000, which aligns with the assumption that claims equal 10% of vehicle value.

Policy Distribution

The portfolio distribution follows the expected structure:

- 3-year policies → largest share
- 2-year policies → moderate share
- 1-year policies → smaller share

- 4-year policies → smallest share

Most Profitable Policy Tenure

3-year policies appear to be the most profitable, with the lowest loss ratio (2.60), indicating the best balance between premium revenue and claim exposure.

Monthly Claim Trends

Monthly claim costs remain relatively stable throughout 2025, averaging around ₹6.4–₹6.7 Crores per month.

However, certain months show abnormal spikes due to defect-related claim clusters.

Portfolio Loss Ratio

The overall portfolio loss ratio is 4.11 (411%), indicating that claim costs significantly exceed premium revenue.

This suggests that the portfolio would be financially unsustainable without adjustments to pricing or risk management strategies.

Potential Future Liability

If all remaining policies eventually file a claim, the insurer could face a maximum potential liability of approximately ₹950.85 Crores.

Future Risk Projection

If claim frequency increases by 5% annually, total claim costs could increase from ₹98.81 Crores to ₹108.69 Crores within two years, significantly impacting profitability.